On the Cost-Function Dependence of the Q-LINK Advantage: Combinatorial Scaling of k-Local Observables Extension 5

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Abstract

The Q-LINK architecture achieves a gradient variance advantage by evaluating the cost function strictly on n data qubits while utilizing an $(n+1)$-qubit extended Hilbert space. This note investigates the stability of this advantage under local cost functions. We first demonstrate that static, unnormalized local operators (e.g., $O = Z_1 Z_2$) perfectly restore trace symmetry, causing the Q-LINK advantage to vanish. To introduce local interaction information while preserving architectural asymmetry, we define a normalized k-local averaged cost operator. By evaluating the Hilbert-Schmidt trace combinatorics, we prove that the relative variance advantage scales as $\mathcal{R}_k(n) = \frac{n+1}{n+1-k}$, demonstrating that highly non-local operators actively amplify the structural Q-LINK advantage.

1 The Collapse of Advantage Under Static Local Costs

Consider a static, unnormalized 2-local cost operator $O_{local} = Z_1 Z_2$. The traceless part is simply $V = Z_1 Z_2$. For a Vanilla circuit on $n+1$ qubits, the squared trace is:

$$Tr(V_{Vanilla}^2) = Tr((Z_1 Z_2 \otimes I^{\otimes n-1})^2) = Tr(I^{\otimes n+1}) = 2N \tag{1}$$

where $N = 2^n$. For the Q-LINK architecture, the operator acts trivially on the messenger: $\tilde{V} = Z_1 Z_2 \otimes I^{\otimes n-2} \otimes I_m$. Its trace is identical:

$$Tr(\tilde{V}^2) = Tr(I^{\otimes n} \otimes I_m) = 2N \tag{2}$$

Because the trace symmetry is perfectly restored ($2N = 2N$), the application of the Master Formula yields an asymptotic variance ratio of strictly 1. The structural advantage of Q-LINK completely evaporates when the cost function lacks extensive normalization.

2 The Averaged k-Local Cost Operator

To incorporate k-local information while preserving the volumetric asymmetry of the measurements, we define the cost function as the normalized global average over all possible k-body Z-interactions.

On the n data qubits of the Q-LINK architecture, the number of such terms is $\binom{n}{k}$. The cost operator is:

$$O_k = I - \frac{1}{\binom{n}{k}} \sum_{|S|=k} \left(\bigotimes_{i \in S} Z_i \right) \tag{3}$$

By Pauli orthogonality, all cross-terms in the squared Hilbert-Schmidt norm of the traceless part $\tilde{V}$ vanish. The trace evaluates to:

$$Tr(\tilde{V}^2) = \frac{1}{\binom{n}{k}^2} \sum_{|S|=k} Tr\left(\left(\bigotimes_{i\in S} Z_i \otimes I_m\right)^2\right) = \frac{1}{\binom{n}{k}^2} \cdot \binom{n}{k} \cdot 2N = \frac{2N}{\binom{n}{k}} \tag{4}$$

For a fair comparison, the equivalent Vanilla circuit operating on $n+1$ qubits averages over $\binom{n+1}{k}$ possible terms:

$$Tr(V_{Vanilla}^2) = \frac{1}{\binom{n+1}{k}^2} \cdot \binom{n+1}{k} \cdot 2N = \frac{2N}{\binom{n+1}{k}} \tag{5}$$

3 Combinatorial Variance Scaling

We compute the asymptotic gradient variance ratio $\mathcal{R}_k(n)$ at Haar saturation by taking the ratio of the traces derived in equations (4) and (5):

$$\mathcal{R}_k(n) = \frac{Tr(\tilde{V}^2)}{Tr(V_{Vanilla}^2)} = \frac{\frac{2N}{\binom{n}{k}}}{\frac{2N}{\binom{n+1}{k}}} = \frac{\binom{n+1}{k}}{\binom{n}{k}} \tag{6}$$

Expanding the binomial coefficients yields:

$$\mathcal{R}_k(n) = \frac{\frac{(n+1)!}{k!(n+1-k)!}}{\frac{n!}{k!(n-k)!}} = \frac{n+1}{n+1-k} \tag{7}$$

4 Conclusion

Equation (7) reveals that the relative Q-LINK ensemble variance advantage is strictly dependent on the locality k of the normalized cost operator.

- For $k = 1$, we recover the baseline linear average scaling: $\mathcal{R}_1(n) = \frac{n+1}{n}$.
- For $k = 2$ (all-to-all pair interactions), the advantage amplifies to $\mathcal{R}_2(n) = \frac{n+1}{n-1}$.

As the locality parameter k approaches the register size n, the combinatorial explosion of measurement sites in the $(n+1)$-qubit Vanilla register severely dilutes its trace norm. The Q-LINK messenger qubit acts as a combinatorial shield against this dilution, actively amplifying the structural trainability advantage for highly non-local cost functions.